

Database Design Document (DDD)

IntelliPlant AI

Database Design Document

Version: 1.0

1. Database Architecture

IntelliPlant AI uses polyglot persistence.

IntelliPlant AI

| PostgreSQL | Neo4j | Qdrant | Redis | MinIO |

2. PostgreSQL Schema

users

Stores users and roles.

```
CREATE TABLE users(  
  id UUID PRIMARY KEY,  
  name VARCHAR(255),  
  email VARCHAR(255) UNIQUE,  
  password_hash TEXT,  
  role VARCHAR(50),  
  status VARCHAR(20),  
  created_at TIMESTAMP  
);
```

documents

```
CREATE TABLE documents(  
  id UUID PRIMARY KEY,  
  filename TEXT,  
  document_type VARCHAR(100),  
  owner_id UUID,  
  storage_path TEXT,  
  status VARCHAR(50),  
  created_at TIMESTAMP  
);
```

chunks

```
CREATE TABLE chunks(  
  id UUID PRIMARY KEY,  
  document_id UUID,  
  chunk_index INT,  
  chunk_text TEXT,  
  embedding_id VARCHAR(255)  
);
```

incidents

```
CREATE TABLE incidents(  
  id UUID PRIMARY KEY,  
  equipment_id VARCHAR(50),  
  failure_type VARCHAR(255),  
  severity VARCHAR(50),  
  description TEXT,  
  incident_date TIMESTAMP  
);
```

inspections

```
CREATE TABLE inspections(  
  id UUID PRIMARY KEY,
```

```
equipment_id VARCHAR(50),
inspector VARCHAR(255),
inspection_date TIMESTAMP,
status VARCHAR(50)
);
```

work_orders

```
CREATE TABLE work_orders(
  id UUID PRIMARY KEY,
  equipment_id VARCHAR(50),
  work_type VARCHAR(100),
  assigned_to VARCHAR(255),
  status VARCHAR(50)
);
```

audit_logs

```
CREATE TABLE audit_logs(
  id UUID PRIMARY KEY,
  user_id UUID,
  action TEXT,
  timestamp TIMESTAMP
);
```

3. Neo4j Schema

Nodes

Equipment

Failure

Incident

Manual

Inspection

Operator

Regulation

WorkOrder

Plant

Relationships

CONNECTED_TO

FAILED_IN

CAUSES

INSPECTED_BY

REQUIRES

DESCRIBED_BY

REFERENCES

SIMILAR_TO

LOCATED_IN

4. Qdrant Collections

Collection

document_chunks

Metadata

```
{  
  "document_id": "",  
  "chunk_text": "",  
  "equipment": "",  
}
```

```
"source": ""  
}
```

Embedding Model

BGE-Large

Dimensions

1024

5. Redis

Stores

- Session state
 - Cache
 - Recent conversations
 - Agent memory
-

6. MinIO

Stores

```
documents/  
manuals/  
images/  
reports/  
inspections/  
work_orders/
```

Conclusion

Polyglot persistence allows IntelliPlant AI to optimize structured, graph, vector, cache, and file storage independently.